4.1 Enzymes

In living cells, most catalysts are protein molecules called enzymes which catalyze numerous biochemical reactions occurring in a cell. Some can also be nucleic acids that behave like enzymes such as ribozymes. They are species of RNA molecules having enzymatic activity. They have a high degree of specificity, that is, they catalyze only one type of reaction, and most act on only one particular substrate. Enzymes of commercial importance are available from plant and animal sources. However, they have certain limitations. The production of enzymes from plants and animals is in short supply and cannot meet the increased demands of the industries. Besides, the industries require a continuous supply of these enzymes. The microbial enzymes faced initial hiccups in their acceptance as the people feared of the enzymes being contaminated by the microbes. However, the microbial enzymes have certain advantages:

1. They can be produced on a large scale by the process of fermentation.
2. If the enzymes are excreted into the medium, they can be isolated easily.
3. A large variety of enzymes can be produced using microbes.
4. The microbes can be genetically engineered to give high yield of enzymes.

The majority of enzymes are now derived from microorganisms followed by plants and animals. Enzymes are especially useful in industrial processes because of their specificity. They act on a certain substrate and yield a certain product. This minimizes the problems of product purification.

Enzymology

The study of enzymes is known as enzymology. Louis Pasteur and others believed that living systems had a "vital force" that allowed them to bypass the laws of nature governing non-living matter as they were unable to reproduce most biochemical reactions in the laboratory. Justus von Liebig and others were of the opinion that biological processes were caused by the action of chemical substances known as *ferments*. The term enzyme (*en* = in; *zyme* = yeast) was coined in 1878 by Wilhelm Kühne. It indicated that something present in the yeast, as opposed to the yeast itself, catalyzes the fermentation reactions. Eduard Buchner reported

that a cell-free yeast extract was able to carry out the synthesis of ethanol from glucose by the following reaction in which 12 enzyme-catalyzed steps are involved:

$$C_6H_{12}O_6 \rightarrow 2CH_3CH_2OH + 2CO_2$$

Since enzymes are proteins, they get destroyed if the temperature increases beyond 40°C. This is the main point of difference between an enzyme and an inorganic catalyst which can function at high temperatures. However, there are some enzymes which function at high temperatures also. For example, those found in thermophilic organisms present in hot sulphur springs and hydrothermal vents, *Taq* polymerase used in polymerase chain reaction. They are functional at 80–90°C and exhibit thermal stability.

Mode of Action of Enzymes

Most of the enzymes are globular proteins. Few are RNA enzymes like ribozyme and ribonuclease P. They are generally hydrophilic in nature. Enzymes act as catalysts by increasing the rates of reactions. Like proteins, enzymes have secondary and tertiary structures. In their tertiary structure, the crevices, clefts or pockets are present that act as active sites (portion of the enzyme molecule which is active and wherein substrate usually binds for efficient biocatalysis). It is the region of the enzyme where catalysis occurs. Catalysis depends on the arrangement of functional groups in the enzyme's active sites which helps in increasing the rates of reactions. Substrates like amino acids bind to the enzyme at this active site. The amino acids present in the active site are usually situated at a distance from each other along the length of the polypeptide chain, but they are brought into close proximity to one another as the polypeptide folds to form the tertiary structure. Amino acid residues lining active site facilitate the bond breakage/formation, group transfer, rearrangement, acceptance/donation of protons, thus facilitate the product formation and accomplish the reaction.

RNA catalysis : (Some species of RNA molecules have enzymatic activity, they are known as ribozymes. They were discovered in the early 1980s. RNA functioned both as a catalyst for some biochemical reactions instead of protein enzymes as well as genetic material. RNA was involved in many important life processes such as metabolism, translation, slicing, etc. A few reactions, such as the cleavage and ligation of phosphodiester bonds required for RNA splicing and the formation of peptide bonds during protein synthesis, are catalyzed by naturally occurring RNAs. Ribosomal RNA that catalyzes the formation of peptide bonds between amino acids is an example of ribozyme.)

Properties of Enzymes

Enzymes do not initiate a chemical reaction, but increase the rate of reaction without getting used up themselves. All the chemical reactions that are controlled by enzymes are reversible. The number of substrate molecules transformed per minute by an enzyme is called turn over number of the enzyme. Enzymes catalyze specific reactions. They do so with great efficiency and with many built-in controls. Three important properties of enzymes are as follows:

1. **Enzymes are highly specific:** Each particular enzyme binds only to specific substrates—the reactant molecules on which the enzyme acts. Of the more than 1000 known enzymes in one's body, each has a characteristic three-dimensional shape with a specific surface configuration, which allows it to recognize and bind to certain substrates. The enzyme specificity is supported by two theories:
   (a) In Lock and Key hypothesis (by Emil Fischer in 1894) (Figure 1), substrate gets fits into the active site of the enzyme as a key fit in a lock. The enzyme specificity is due to the complementary shapes of enzyme and the substrate.

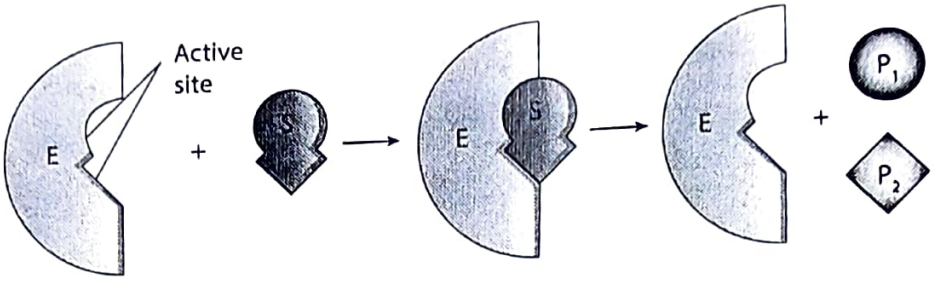


Figure 1 Lock and Key hypothesis.

(b) In 1958, Daniel Koshland suggested the Induced-fit model (Figure 2), where the active site changes its shape to fit around the substrate once the substrate enters the active site. The substrate induces a change in the enzyme shape to fit the shape of the substrate.

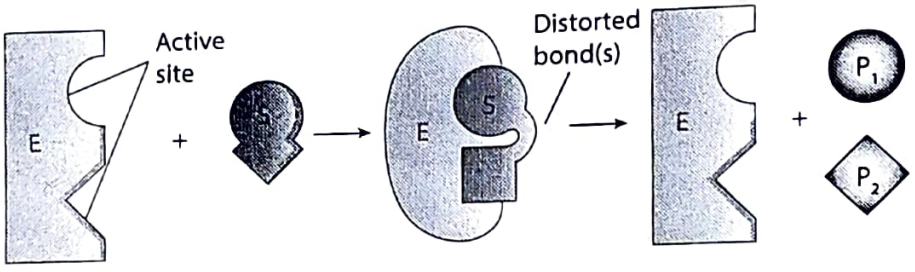


Figure 2 Induced-fit model.

2. **Enzymes are very efficient:** Under optimal conditions; enzymes can catalyze reactions at rates 100 million to 10 billion times more rapid than those of similar reactions occurring without enzymes.
3. **Enzymes are subject to a variety of cellular controls:** Their rate of synthesis and their concentration at any given time are under the control of a cell's genes. Substances within the cell may either enhance or inhibit the activity of a given enzyme.

Chemical Reactions

In a chemical reaction, the bonds are broken and new bonds are formed. A chemical reaction can be inorganic or organic. The amount of product formed in a given time determines the rate of reaction of physical and chemical processes. It is given by the following equation:

$$\text{Rate} = \frac{\mathrm{d}[P]}{\mathrm{d}t}$$

where, $[P]$ = product formed and t = time.

If a catalyst is added to a reaction, it increases the rate many times higher than those that are not catalyzed. For example, when carbon dioxide reacts with water to form carbonic acid without catalyst, 200 molecules of carbonic acid are formed per hour. However, if the enzyme carbonic anhydrase is present, 600,000 molecules are present per second and the rate of reaction increases by ten million times. Its reaction rate is one of the fastest of all the enzymes.

High Rates of Chemical Conversions and Nature of Enzyme Action

Enzymes help bring the substrates together in the proper orientation so that the reaction can occur. Figure 3 depicts how an enzyme works:

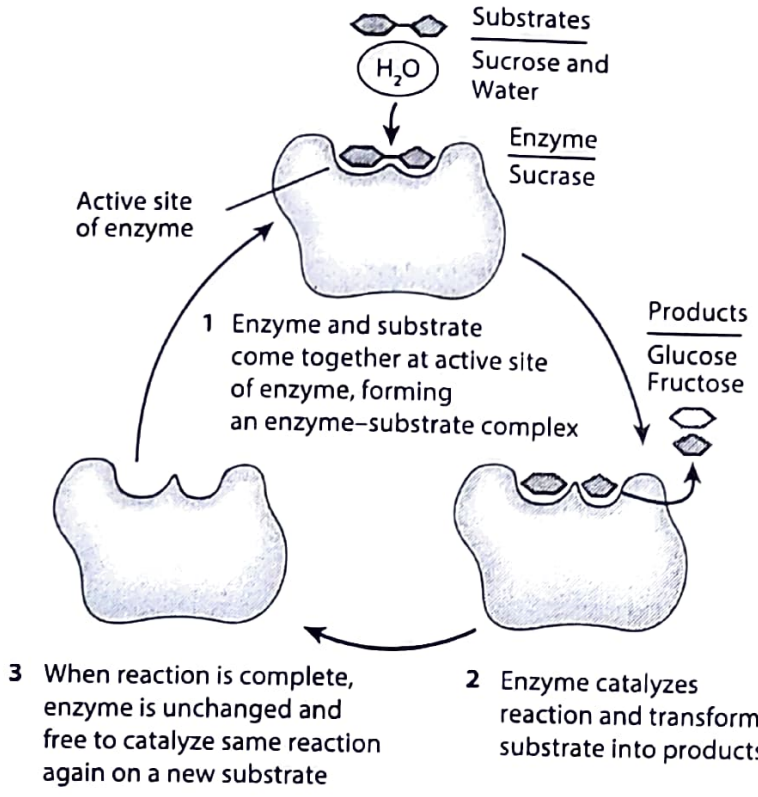


Figure 3 Mechanism of enzyme action.

The three main steps in an enzyme-catalyzed reaction are:

1. **Formation of ES complex:** The substrates make contact with the active site on the surface of the enzyme molecule, forming a temporary intermediate compound called the enzyme–substrate (ES) complex. The binding of substrate to enzyme is accomplished by non-covalent interactions such as ionic bonds, hydrogen bonds and hydrophobic interactions.
2. **Transformation of substrate molecules:** The substrate molecules are transformed by the rearrangement of existing atoms, the breakdown of the substrate molecule, or the combination of several substrate molecules into the products of the reaction. Here the products are two monosaccharides: glucose and fructose.
3. **Product formation:** As the transition state is converted to products, the affinity of the enzyme for the bound molecule(s) decreases as the products are unable to bind to the buttressing group, and get expelled. After the reaction is completed and the reaction products move away from the enzyme, the unchanged enzyme is free to attach to other substrate molecules.

Activation energy is the minimal kinetic energy needed for a reactant to undergo a chemical reaction. It is the difference in energy between the substrate and the transition state. Figure 4 shows the activation energies of catalyzed and uncatalyzed reactions demonstrated by the height of the curves. The *y*-axis shows the potential energy, while the *x*-axis represents the course of reaction. The substrate has to cross the energy barrier and has to reach up to the transition state which is at high energy level. This can be achieved through precipitation of reactants; weak bond formation between reactants and enzymes; increasing the proximity of reactants to active sites on the enzymes; weakening the bonds of reactants through attack by electrophiles or nucleophiles, etc.

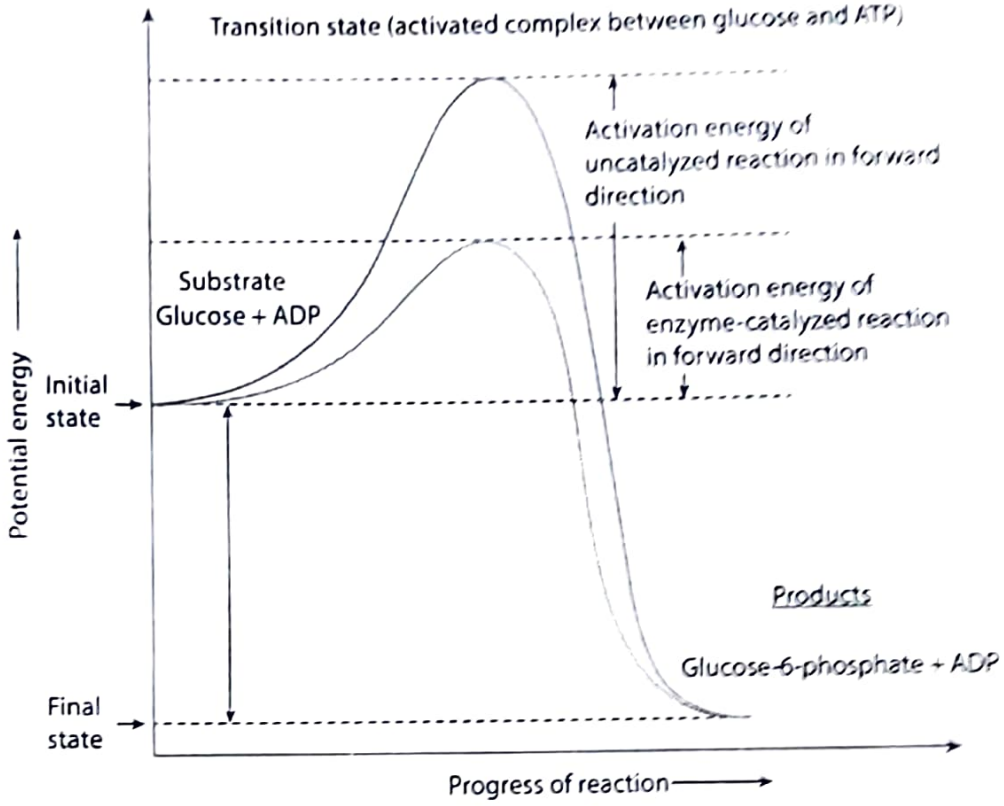


Figure 4 Activation energy and enzymatic reactions.

Factors Affecting Enzyme Activity

The catalytic activity of an enzyme is revealed by studying its kinetics or the rate at which the enzyme catalyzes a reaction. There are many factors that affect the activity of an enzyme and affect the rate of reaction. Some of these are discussed below:

pH and Temperature

Every enzyme operates best at an optimal pH and temperature. Below or above these optimal values, the activity is lowered. These values are different for different enzymes.

Concentration of Substrate

Like pH and temperature, the concentration of substrate also affects enzyme catalysis. When the substrate concentration is low, at any given point in time, the number of collisions between enzyme molecules and substrate is less. When the substrate concentration is high, the enzymes collide with substrate molecules more rapidly than they can be converted to product. With increase in substrate concentration, the enzyme reaches a state of saturation.

In 1913, Leonor Michaelis and Maud Menten derived a mathematical relationship between substrate concentration and the velocity of enzyme reactions. They measured the amount of product formed (or substrate consumed) in a given time period and gave the following equation, also known as Michaelis–Menten equation.

$$V = V_{max} \frac{[S]}{[S] + K_m}$$

where, V = initial velocity of the reaction; V_{max} = maximum rate of reaction; K_M = Michaelis-Menten constant and $[S]$ = substrate concentration.

According to the equation, when the substrate concentration $[S]$ is set at a value equivalent to K_M, then the velocity of the reaction (V) becomes equal to $V_{max}/2$, or one-half the maximal velocity. Thus, $K_M = [S]$, when $V = V_{max}/2$. Michaelis constant (K_M) is the substrate concentration when the reaction velocity is one-half of Vmax. It is constant for a given enzyme and is independent of substrate or enzyme concentration.

With increase in substrate concentration, the enzyme reaches a state of saturation. This is because all the substrate molecules have been used up and there are no free enzyme molecules left to bind to the additional substrate. At this point, the initial velocity is termed the maximal velocity (V_{max}). The catalytic activity of an enzyme is measured by its turnover number. The turnover number is the maximum number of molecules of substrate that can be converted into product by one enzyme molecule per unit time.

Presence of Specific Chemicals—Enzyme Inhibition

Enzyme inhibitors are molecules that bind to an enzyme and decrease its activity. All cells depend on inhibitors to regulate the activity of other enzymes. These can be divided into two types: reversible and irreversible. Reversible inhibitors can further be divided into competitive and non-competitive.

Irreversible inhibitors strongly bind to an enzyme, usually by forming a covalent bond with its amino acid residues. Reversible inhibitors bind loosely to an enzyme, and thus can be readily displaced. These are of two types (Figure 5):

1. Competitive inhibitors compete directly with a substrate for an enzyme's substrate binding site. They are structurally similar to the substrate which enables them to specifically bind to the active site. For example, malonate is a competitive inhibitor of succinate dehydrogenase, an enzyme that converts succinate to fumarate.
2. Non-competitive inhibitors do not compete with the substrate for the same binding site of the enzyme; rather, the inhibitor binds to a site other than the enzyme's active site. This changes the conformation of the active site in such a manner that the substrate can no longer bind to the active site. This type of inhibition depends only on the inhibitor concentration, and is independent of the concentration of the substrate. The inactivation of cytochrome oxidase by cyanide is an example of non-competitive inhibition.

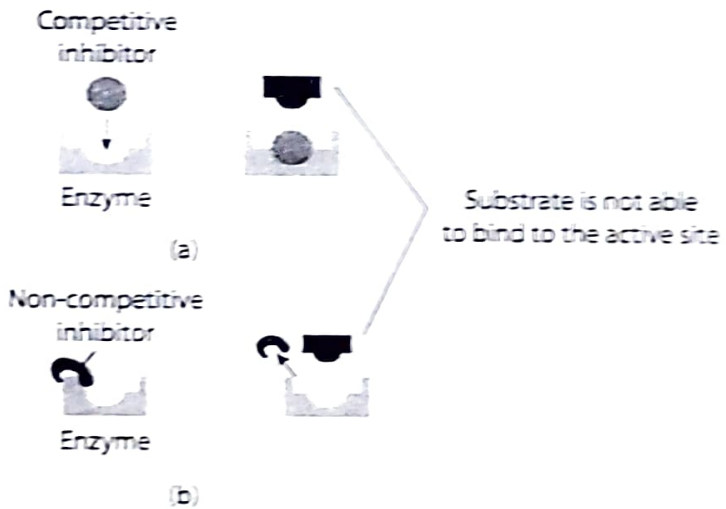


Figure 5 Enzyme inhibition. (a) Competitive inhibition and (b) non-competitive inhibition.

Allosteric Enzymes

An enzyme's catalytic activity can be modulated. This can be done either negatively or positively through conformational changes influencing the enzyme–substrate affinity or by changing the turnover number.

A substance that binds to a site other than the active site and influences the active site either negatively or positively is called a modulator or effector molecule. The effector molecules that enhance the enzymes activity are called allosteric activators. When it causes a decrease in enzyme activity, the effector molecules are called allosteric inhibitors.

Feedback inhibition takes place by a product molecule which inhibits an early step in a reaction sequence in its own biosynthesis. This happens when a product accumulates beyond its optimal amount, it binds to the allosteric site of the enzyme catalysis its synthesis, thus altering the conformation of the enzyme such that it no longer binds to the substrate.

4.2 Classification and Nomenclature of Enzymes

Enzymes are commonly named by appending the suffix -ase to the name of the enzyme's substrate (urease catalyzes the hydrolysis of urea) or to the word describing the enzyme action (alcohol dehydrogenase catalyzes the oxidation of primary and secondary alcohols to their corresponding aldehydes and ketones by removing hydrogen). A number of enzymes are known today. In the beginning, there were no systematic rules for naming enzymes which resulted in using two different names for the same enzyme, thereby creating confusion. To avoid these problems, the International Union of Biochemistry and Molecular Biology (IUBMB) adopted the systematic functional classification and nomenclature of enzymes.

On the basis of nature of the reactions that the enzymes catalyze, they are divided into 6 classes (Table 1).

Table 1 Enzyme Classification According to Reaction Type

Classification	*Type of Reaction Catalyzed*	*Examples*
Oxidoreductases/ dehydrogenases	Oxidation-reduction reactions	Nitrate reductase, succinate dehydrogenase
Transferases	Transfer of functional groups, example, amino group from glutamate to pyruvate in alanine synthesis	CoA transferase, N-acetyl transferase
Hydrolases	Hydrolysis reactions of ether, ester, C–C, C–X, peptide, glucosidic bonds	Amylase, lactase, maltase
Lyases	Group elimination to form double bonds	Histidine decarboxylase, aldolase
Isomerases	Isomerization to form optical, positional or geometrical isomers	Alanine racemase, tartrate epimerase
Ligases	Bond formation of C–O, P–O, C–S, C–N coupled with ATP hydrolysis	Pyruvate carboxylase

Few examples of enzyme and their functions are:

1. **Lactate dehydrogenase:** It belongs to the class of dehydrogenase. It facilitates the interconversion between lactate and pyruvic acid with the reduction of nicotinamide adenine dinucleotide (NAD^+ to NADH).
2. **Nucleoside monophosphate kinase:** It belongs to the class of transferase enzyme as it facilitates the transfer of the phosphoryl group from ATP to nucleoside monophosphate. It catalyzes the following chemical reaction:

$$\text{ATP} + \text{Nucleoside phsopshate} \rightleftharpoons \text{ADP} + \text{Nucleoside diphosphate}$$

3. **Protease:** It belongs to the class of hydrolase. They are involved in breakdown of proteins into shorter peptides and amino acids. These enzymes use water to break down the chemical bond.
4. **Pyruvate decarboxylase:** It belongs to the class of lyase. It catalyzes the decarboxylation of pyruvic acid to acetaldehyde and carbon dioxide. The cofactor TPP (Thiamine pyrophosphate) acts as prosthetic group for the reaction.
5. **Alanine racemase:** It belongs to the class of isomerase. It catalyzes the conversion of L-alanine into its isomeric (mirror-image) form, D-alanine.
6. **DNA ligase:** It belongs to the class of ligase. It facilitates the joining of DNA strands together by catalyzing the formation of a phosphodiester bond.
7. **Restriction enzymes:** Restriction enzymes are also called molecular scissors because these enzymes recognize the base sequence at specific palindrome sites in DNA duplex and cut its strands, by cleaving phosphodiester bonds. They belong to the class of enzymes called nucleases. The first restriction enzyme that recognized and cut DNA at a specific in 1968 by H.O. Smith, K.W. Wilcox and T.J. Kelly. They isolated and characterized an enzyme from *Haemophilus influenzae* and named it *Hind* II. This enzyme cuts the DNA molecule at a particular site within a specific sequence of six base pairs.

   5′GT (pyrimidine: T or C) ↓ (purine: A or G) A C 3′
   3′CA (purine: A or G) ↓ (pyrimidine: T or C) T G 5′

   They named this sequence as recognition sequence and showed that *Hind*II always cut between the third and fourth base pairs. Other examples of restriction endonucleases are *Eco*RI found in *Escherichia coli* type-*R* and *Hpa*I from *Haemophilus parainfluenzae*. More than 900 different types of restriction endonucleases have been isolated so far from more than 230 strains of bacteria.

4.3 Co-Factors

Enzymes can be simply made up of one or more chains of polypeptides (like urease, pepsin) or may be conjugated with a co-factor. Co-factor is the non-protein portion which is generally associated with the protein part to make it catalytically active. It is of three types: metal ions, co-enzyme and prosthetic group (Figure 6). Apoenzyme is the protein portion of the enzyme without a co-factor. Holoenzyme is a catalytically active apoenzyme–co-factor complex.

$$\text{Apoenzyme (inactive)} + \text{Co-factor} \rightleftharpoons \text{Holoenzyme (active)}$$

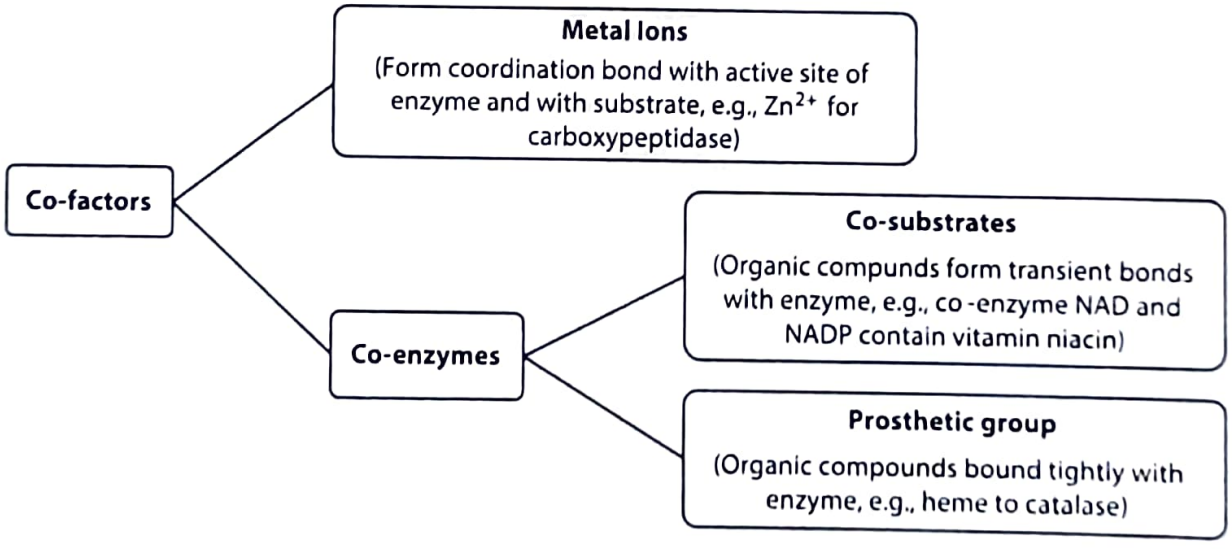


Figure 6 Types of co-factors.

4.4 Importance of Enzymes

1. They act as biocatalysts synthesized by living cells and take part in various biochemical reactions.
2. Due to their unique properties of specificity and flexibility, they act as dynamic catalysts and affect the reaction based on the desired conditions.
3. Cofactors act as co-substrates and bring about prosthetic groups, coenzymes and metal ions bind to the active sites of enzymes to maintain the shape of the site so that a stable enzyme–substrate complex is formed.
4. Inhibitors are used to study the properties of enzymes and are used in form of drugs, antibiotics, or pesticides.